

FROM SINGLE-CELL MODELING TO LARGE-SCALE NETWORK DYNAMICS WITH NEST SIMULATOR

Charl Linssen | October 14th, 2026 | EBRAINS Virtual Users' Week

EBRAINS

RWTH AACHEN UNIVERSITY

JÜLICH Forschungszentrum
Member of the Helmholtz Association

Welcome!

09:30	Welcome and overview
09:35	Introduction to NEST Simulator
09:45	Creating and simulating a balanced network
10:15	Short break
10:30	Introduction to NESTML
10:45	Custom spike-frequency adaptation models
11:00	Custom synaptic plasticity models
11:30	Closing

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Prerequisites: Python programming and JupyterLab

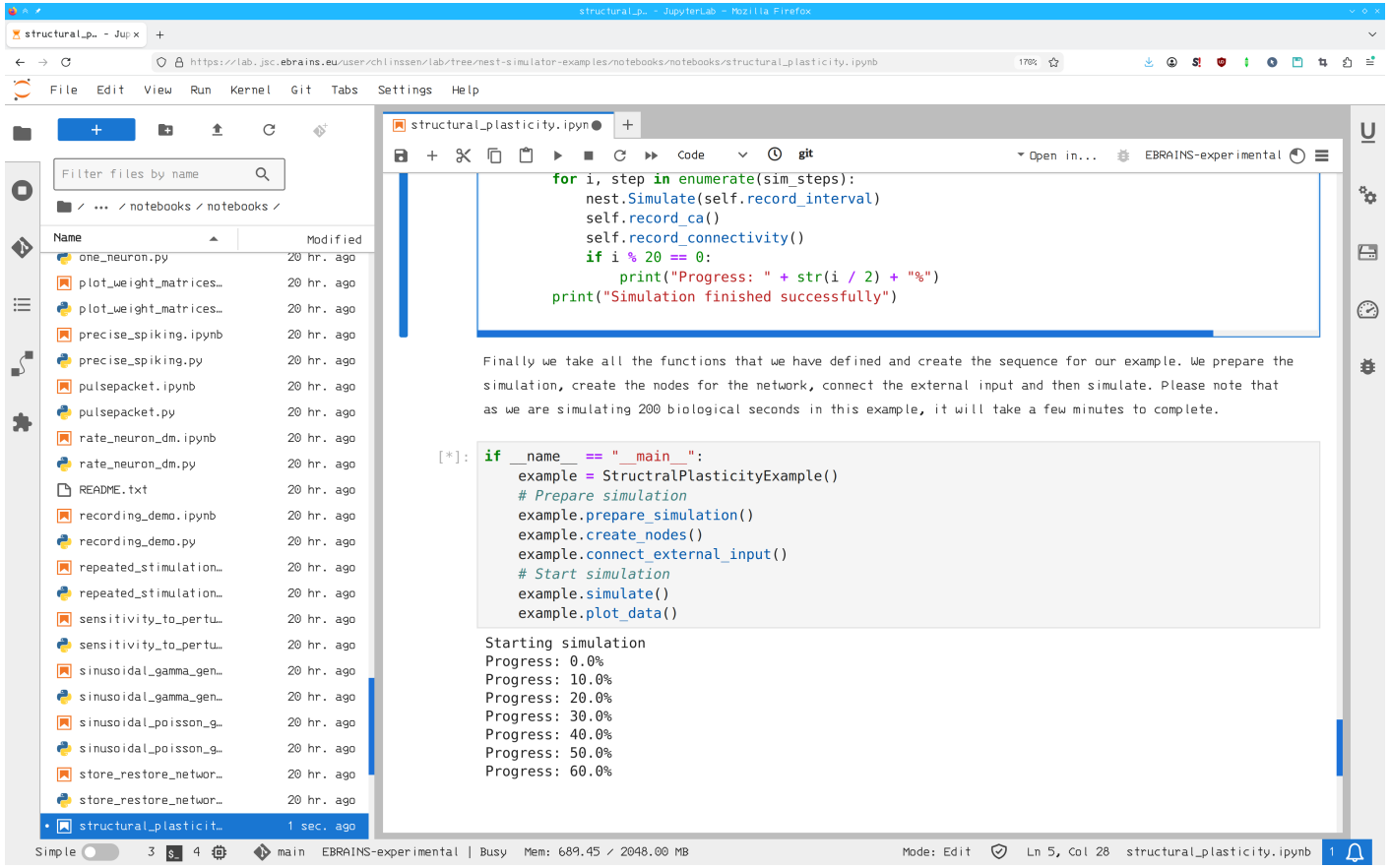
JupyterLab is an interactive graphical user interface (GUI) for the Python programming language.

Usage tips:

- Shift+Return runs a single cell.
- Use the ⏩ icon in the toolbar to run all cells.

For an advanced JupyerLab tutorial, see e.g. https://realpython.com/using-jupyterlab/

Code will run inside an EBRAINS VM with 2 GB RAM. Files stored in your VM home directory can disappear—save them locally for safekeeping!



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Running the balanced network tutorial notebook

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Running the balanced network tutorial notebook

① Use your browser to go to the URL:

https://lab.ebrains.eu/

Select JSC as execution site.

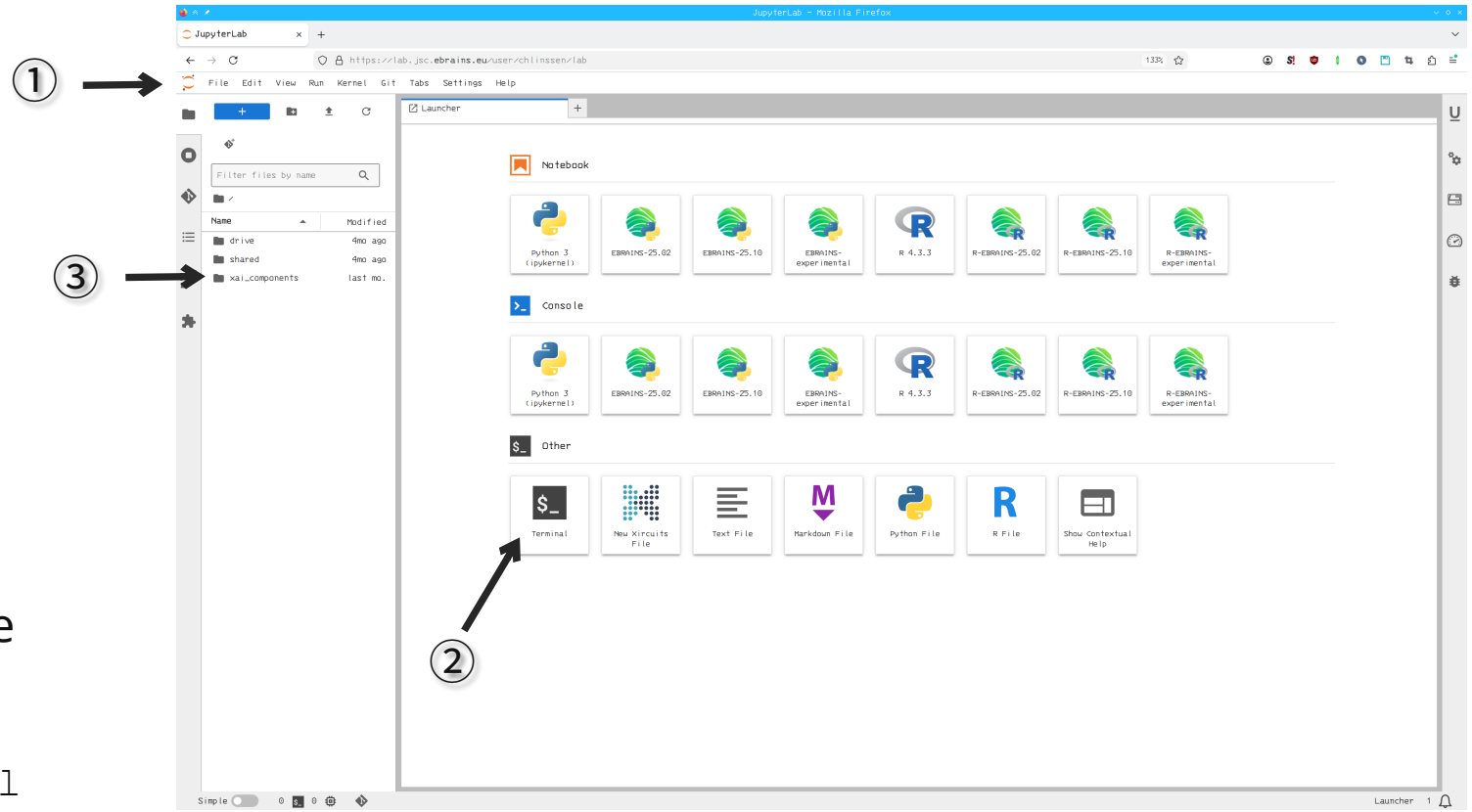
② Open a terminal and run:

```
git clone
https://github.com/clinssen/NEST-
workshop
```

③ The notebooks we will cover today are in:

```
materials/nest/nest_intro_and_brunel
```

④ Select the “EBRAINS-25.10” kernel



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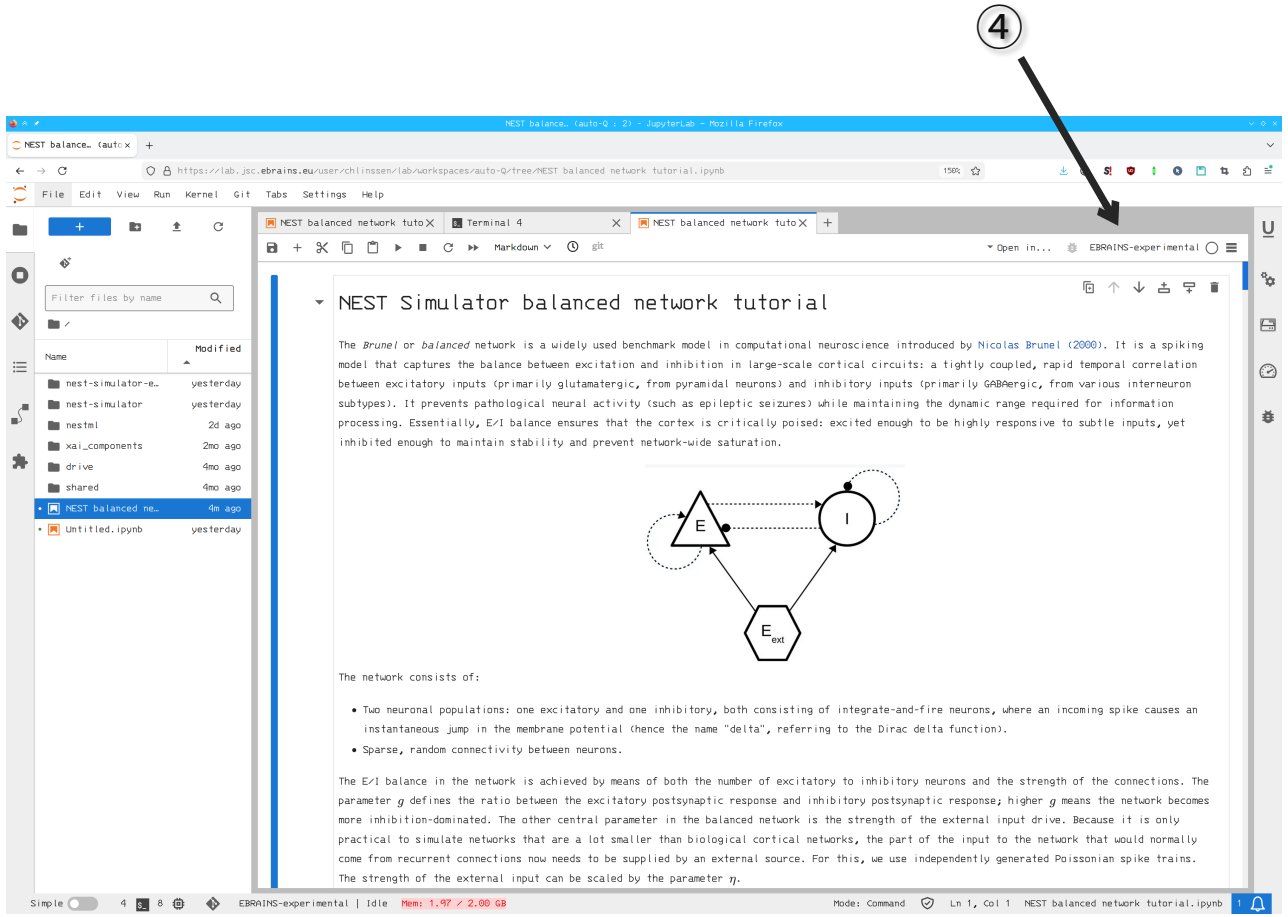
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Tutorial materials for NEST Simulator

- Check out other tutorials in the “materials” subdirectories of the repository:

  https://github.com/clinssen/NEST-workshop

- Central place for NEST Simulator documentation:

  https://nest-simulator.readthedocs.io/

  See especially the “examples” section.

- Cortical microcircuit model:

  https://microcircuit-pd14-model.readthedocs.io/en/latest/

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Creating custom neuron and synapse models using NESTML

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Running NESTML

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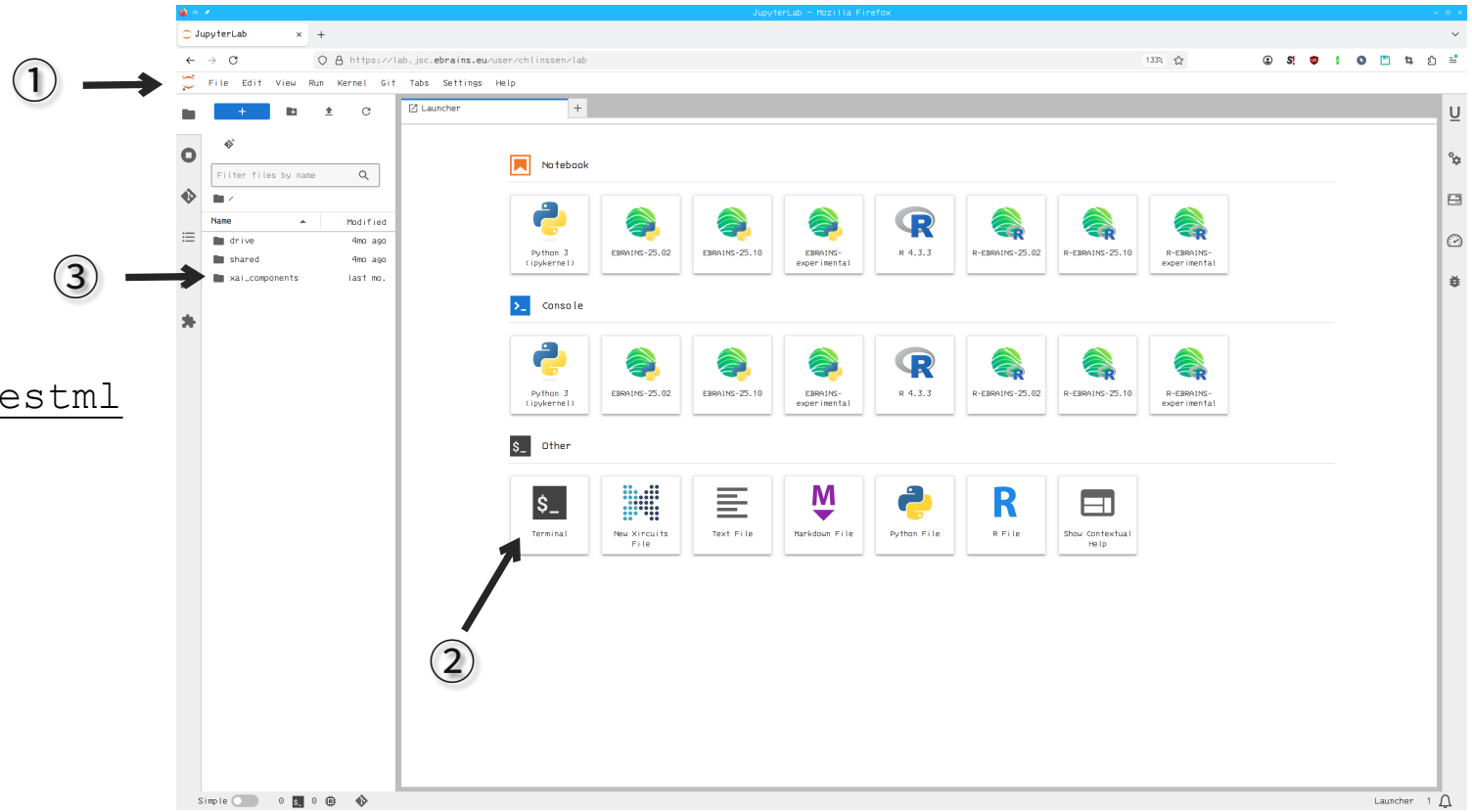
```
git clone https://github.com/nest/nestml
cd nestml
git checkout v8.3.0
```

③ Go back to the start page and open a tutorial notebook:

```
nestml/doc/tutorials/...
```

We will start with the “Spike frequency adaptation” notebook.

④ Select the “EBRAINS Experimental” kernel



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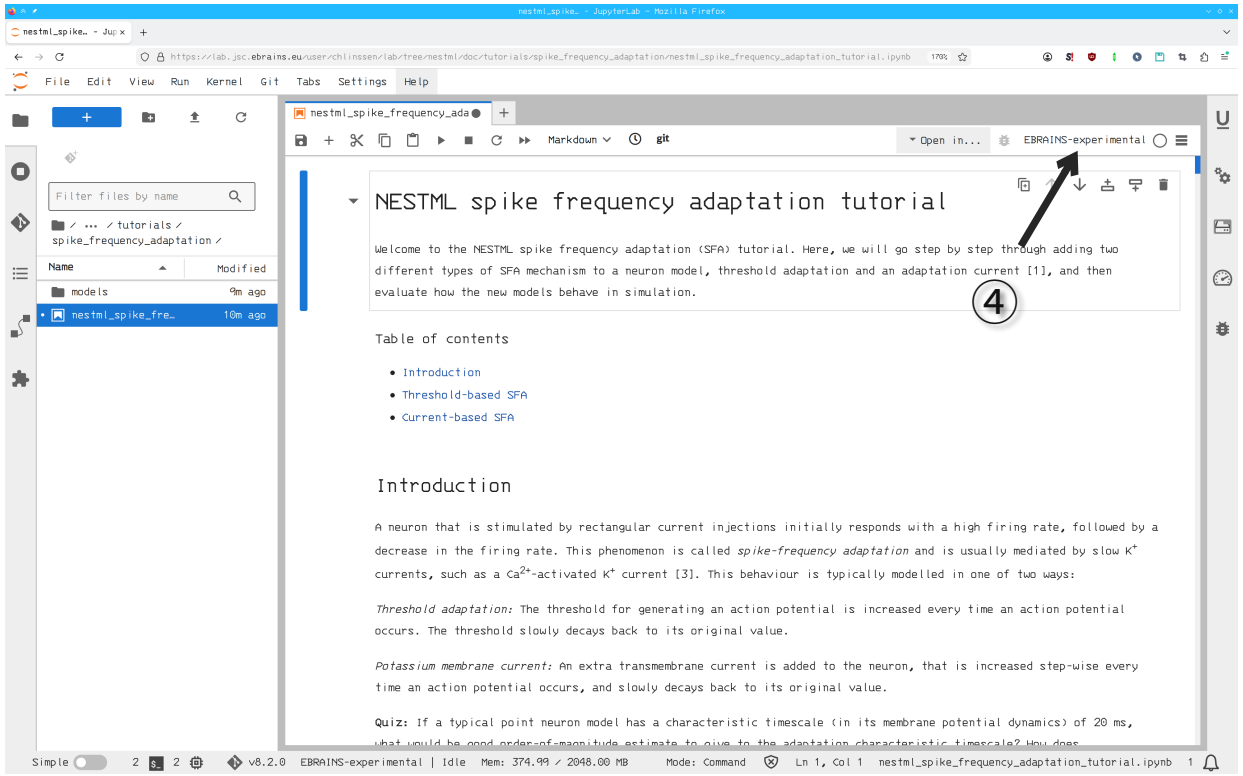
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Tutorial materials for NESTML

- Central place for NESTML documentation:

  https://nestml.readthedocs.io/

- All available tutorials are in the `nestml/doc/tutorials/` directory that you git cloned.

Thank you for your participation!

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https://www.surveymonkey.com/r/L5L72YB

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